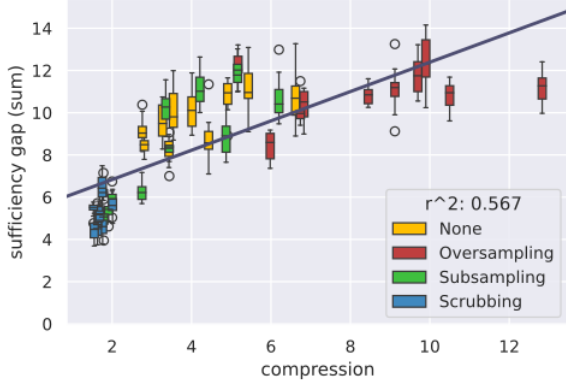
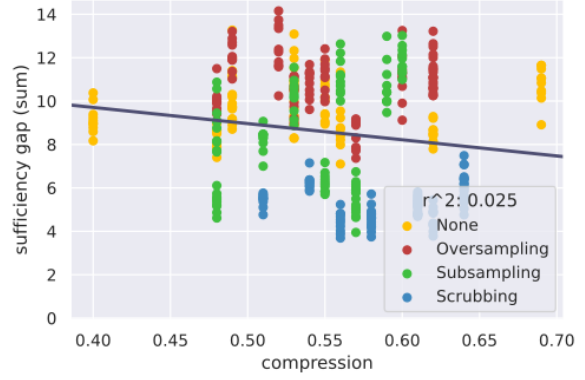




(a) Intrinsic metric was measured on the test set of occupation prediction, figure reproduced from Orgad et al. (2022).



(b) Intrinsic metric was measured on Winobias (Zhao et al., 2018a).

Figure 3: Correlation between an intrinsic metric (compression) and an extrinsic metric (sufficiency gap sums), for various models trained on occupation prediction task. “None” was trained on the original dataset, “Oversampling” was trained on an oversampled dataset, “Subsampling” was trained on a subsampled dataset and “Scrubbing” was trained on a scrubbed dataset (explicit gender words like “he” and “she” were removed).

4.2 Case study: measuring intrinsic bias on two different datasets

It is critical to consider the impact of the data used when measuring intrinsic bias metrics on a language model. Previous work (Goldfarb-Tarrant et al., 2021; Cao et al., 2022; Orgad et al., 2022) inspected the correlations between extrinsic and intrinsic gender bias metrics. Some did not find correlations, while others did in some cases. However, correlations do not solely depend on the model used for bias measurement, but also on the dataset used to measure the intrinsic metric.

Our experiment analyzes the behavior of the same metric on different datasets. We again follow Orgad et al. (2022), who probed for the amount of gender information extractable from the model’s internal representations. This is quantified by *compression* (Voita and Titov, 2020), where a higher compression indicates greater bias extractability. Orgad et al. found that this metric correlates strongly with various extrinsic metrics. An example of this correlation is shown in Figure 3a on the Bias in Bios task with models debiased with various strategies. The correlation is high ($r^2 = 0.567$).

In their experiment, the intrinsic metric was measured on the same dataset as the extrinsic one. We repeat the correlation tests, but this time measure the intrinsic metric on a different dataset, the Winobias dataset. The results (Figure 3b) clearly show that there is no correlation between extrinsic and intrinsic metrics in this case ($r^2 = 0.025$).

Hence, we conclude that the dataset used to measure intrinsic bias impacts the results significantly.

To reliably reflect the biases that the model has acquired, it should be closely related to the task that the model was trained on. In our experiment, when intrinsic and extrinsic metrics were not measured on the same dataset, no correlation was detected. This is the case for all metrics on this task from Orgad et al. (2022); see Appendix 3. As discussed in §3, the same intrinsic metrics can be evaluated across a variety of datasets. Even so, some intrinsic metrics were originally defined to be measured on different datasets than the task dataset, such as those defined on templates (Table 1).

5 Different Metrics Cover Different Aspects of Bias

In this section, we explore how the choice of bias metrics influences results. Although extrinsic bias metrics are useful in defining harms caused by a gender-biased system, we find that most studies on gender bias use only intrinsic metrics to support their claims. We surveyed a representative list of papers presenting bias mitigation techniques that appeared in the survey by Stanczak and Augenstein (2021), as well as recent papers from the past year. In total, we examined 36 papers. Many papers do not measure extrinsic metrics. Even when downstream tasks are measured, only a very small subset of metrics (three or less) is typically measured, as shown in Figure 4. Furthermore, in these studies, typically no explanation is provided for choosing a particular metric.

The exceptions are de Vassimon Manela et al. (2021) and Orgad et al. (2022), who measured six

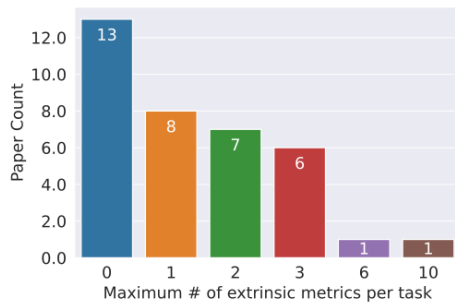


Figure 4: The number of extrinsic metrics measured in the papers we reviewed.

and nine or ten metrics on downstream tasks, respectively. [Orgad et al.](#) showed that different extrinsic metrics behave differently under various debiasing methods. Additionally, in §4 we saw that sub-sampling the test set increased one bias metric and decreased others, which would not have been evident had we only measured a small number of metrics. Measuring multiple metrics is also important for evaluating debiasing. When [Kaneko and Bollegala](#) compared their proposed debiasing method to that of [Dev et al. \(2020a\)](#), the new method outperformed the old one on two of the three metrics.

As the examples above illustrate, different extrinsic metrics are not necessarily consistent with one another. Furthermore, it is possible to measure more extrinsic metrics, although it is rarely done. When it is not feasible to measure multiple metrics, one should at least justify why a particular metric was chosen. In a CV-filtering system, for example, one might be more forgiving of FPR gaps than of TPR gaps, as the latter leaves out valid candidates for the job in one gender more than the other. However, more extrinsic metrics are likely to provide a more reliable picture of a model’s bias.

6 Conclusion and Proposed Guidelines

The issues described in this paper concern the instabilities and vagueness of gender bias metrics in NLP. Since bias measurements are integral to bias research, this instability limits progress. We now provide several guidelines for improving the reliability of gender bias research in NLP.

Focus on downstream tasks and extrinsic metrics. Extrinsic metrics are helpful in motivating bias mitigation (§2). However, few datasets can be used to quantify extrinsic metrics, especially extrinsic performance metrics, which require gold labels (§3). More effort should be devoted to collecting datasets with extrinsic bias assessments, from more

diverse domains and downstream tasks.

Stabilize the metric or the dataset. Both the metrics and the datasets could have significant effects over the results: The same dataset can be used to measure many metrics and yield different conclusions (§4), and the same metric can be measured on different datasets that lead to different results and different conclusions (§5). If one wishes to measure gender bias in an NLP system, it is better to hold one of these variables fixed: for example, to focus on a single metric and measure it on a set of datasets. Of course, this can be repeated for other metrics as well. This will produce much richer, more consistent, and more convincing results.

Neutralize dataset noise. As a result of altering a dataset’s composition, we observed very different results (§4). This is caused by the way various fairness metrics are defined and computed on the entire dataset. To ensure a more reliable evaluation, we recommend normalizing a dataset when using it for evaluation. In the case of occupation prediction, normalization can be obtained by balancing the test set. In other cases it could be by anonymizing the test set, removing harmful words, etc., depending on the specific scenario.

Motivate the choice of specific metrics, or measure many. Most work measures only a few metrics (§5). A comprehensive experiment, such as to prove the efficacy of a new debiasing method, is more reliable if many metrics are measured. In some situations, a particular metric may be of interest; in this case one should carefully justify the choice of metric and the harm that is caused when the metric indicates bias. The motivation for debiasing this metric then follows naturally.

Define the motivation for debiasing through bias metrics. [Blodgett et al. \(2020\)](#) found that papers’ motivations are “often vague, inconsistent, and lacking in normative reasoning”. We propose to describe the motivations through the gender bias metrics chosen for the study: define what is the harm measured by a specific metric, what is the behavior of a desired versus a biased system, and how the metric measures it. This is where extrinsic metrics will be particularly useful.

We believe that following these guidelines will enhance clarity and comparability of results, contributing to the advancement of the field.

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